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Double scheduling in virtualized environments

67–85 minutes

reformule : In the context of virtualization, another example of the semantic gap is the double scheduling of processes. The HK schedules vCPUs that are used by the GK to schedule processes within the VM. Here, the semantic gap manifests at the level of vCPUs, which are perceived by the HK as processes, but by the GK as actual CPUs. This leads to double scheduling, as the HK first schedules a vCPU, and then the GK schedules a process (process inside the VM) on that vCPU. The goal of this internship is to measure and remedy this double scheduling.

Apr 5

Virtualization introduces another manifestation of the semantic gap through the double scheduling of processes. The host kernel (HK) schedules vCPUs, which the guest kernel (GK) in turn uses to schedule its own processes within the virtual machine. The semantic gap arises because the HK treats vCPUs as ordinary processes, whereas the GK considers them to be real, physical CPUs. This discrepancy results in two layered scheduling decisions: the HK first decides when a vCPU may run, and then the GK assigns a process to that vCPU, unaware of the underlying scheduling constraints. The objective of this internship is to quantify the impact of this double scheduling and to propose solutions to mitigate it.

question, the HK considers each VCPU as a process right ?

Apr 5

Yes, exactly. From the host kernel's perspective, each vCPU is just a regular process (or more precisely, a thread). The HK has no awareness that these processes actually represent virtual processors running an entire guest operating system. It schedules them the same way it would schedule any other process on the physical CPUs, using its standard scheduling policies (e.g., CFS on Linux).

That's precisely where the semantic gap lies: the HK sees a simple process, while in reality that process carries the weight of an entire CPU for the guest, with its own queue of processes waiting to be scheduled by the GK.

pdf

This is to be written after the intro right ?

Apr 6

Typically yes, but not immediately after. The standard structure of a research paper is usually: Introduction → Related Work / Background → Methodology → Results → Discussion → Conclusion.

So the methodology comes after you've set up the context (intro) and reviewed existing work (related work/background), but before you present your results. Are you working on writing the methodology section for your internship on double scheduling?

ok, resume the document so i can read and start writing

Apr 6

Here's a summary of the key takeaways from the slides:

What the method section is for: It explains how you achieved your results, and allows others to replicate or extend your work.

Key elements to include (not all mandatory, but Procedures and Materials almost always are): overview of the experiment, population/sample, location, restrictions, sampling technique, procedures (algorithm/workflow), materials (hardware), variables, and statistical treatment.

How to organize it: Use either a chronological organization (stages/phases) or a spatial organization (describing components/objects). For your internship on double scheduling, a chronological approach probably fits well (e.g., first we measured, then we analyzed, then we applied a fix).

Use descriptive headings. Don't write "Phase 1, Phase 2." Instead, write something like "Measuring Double Scheduling Overhead" or "Implementing a Scheduling-Aware vCPU Policy." The reader should understand the content from the heading alone.

Write an overview paragraph at the beginning of the section that summarizes the whole method before diving into subsections.

Language rules:

- Use the **simple past tense** to describe what you did ("Experiments were conducted...", "We measured...")
- Use **passive voice** to depersonalize and emphasize the procedure, but alternate with active voice to keep the text readable
- Include enough detail for someone to replicate your work

Want help drafting an outline for your methodology section based on your double scheduling topic?

if what i did is reading a paper, working on an M2 lab to understand virtualization and start reading the code, what should be my titles

Apr 6

2 / 2

Careful here — the methodology section describes how you carry out your research to obtain results, not your learning process. Reading papers and doing labs are preparatory steps, not methodology.

That said, if you're writing an early-stage internship report (like a progress report), you could frame these activities as part of your approach. Something like:

Section title: "Understanding and Characterizing Double Scheduling in Virtualized Environments"

Then for subsections:

- **"Analysis of Existing Approaches"** → covers the paper you read and what you learned from it (what solutions exist, what gaps remain)
- **"Experimental Familiarization with the Virtualization Stack"** → covers the M2 lab, what you set up, what you understood about how the HK/GK interaction works in practice
- **"Source Code Analysis of the Host Scheduler"** → covers your code reading, which parts of the kernel you explored, what you identified as relevant to double scheduling

This way you're not just saying "I read stuff," you're framing it as investigative steps toward your research goal. Each subsection should explain *why* you did it and *what you learned* that feeds into the next phase of your work.

Does this match the stage you're at, or have you also started running experiments or modifying code?

explain each

Apr 6

Here's what each element means, applied to your double scheduling internship context:

1. Overview — A short paragraph summarizing your entire methodology. "We studied double scheduling by first analyzing existing literature, then familiarizing ourselves with the virtualization stack, and finally examining the kernel source code."

2. Location — Where the work was carried out. In computer science, this typically means the platform or environment: which machine, which OS, which hypervisor (e.g., KVM on Linux 6.x, running on a server with X cores).

3. Limiting conditions — Constraints or boundaries of your study. For example: "We focused only on KVM-based virtualization," or "Experiments were limited to CPU-bound workloads."

4. Samples description — What you observed or tested. In your case, this could be the workloads, the VMs, the number of vCPUs configured, etc.

5. Sampling technique/procedure — How you chose your samples. Why these specific workloads? Why this number of vCPUs? How did you decide what to measure?

6–8. Procedure/Materials — The core of your method. What you actually did step by step, and what tools/hardware you used. Multiple entries because there are usually several procedural steps (e.g., setting up VMs, running benchmarks, collecting traces).

9. Variables/sampling technique — What you measured (dependent variables like latency, CPU idle time) and what you varied (independent variables like number of vCPUs, workload type).

10. Statistical analysis — How you processed the data.

Averages, confidence intervals, number of repetitions, etc.

Want me to help you map your current work onto these elements?

how does that translate to my case

Apr 6

Here's how each element could translate to your internship:

1. Overview — "This work aims to measure and mitigate the double scheduling problem that arises when a host kernel schedules vCPUs that are in turn used by the guest kernel to schedule its own processes. Our approach consisted of three phases: studying existing work, understanding the virtualization stack, and analyzing the relevant kernel code."

2. Location — The lab/team you're working in (e.g., Inria Grenoble, LIG), and the machine(s) you're using. Which Linux kernel version, which hypervisor (KVM/QEMU), what hardware specs (CPU model, number of cores, RAM).

3. Limiting conditions — You're focusing on KVM only (not Xen or VMware). Maybe only Linux as both host and guest. Possibly only CPU scheduling (not I/O or memory). Only a specific kernel version.

4. Samples description — The VM configurations you're studying: how many vCPUs, how much memory, what guest OS. The workloads you plan to run inside the VMs.

5. Sampling technique — Why you chose those specific configurations. For instance, testing with 1, 2, 4, 8 vCPUs to see how overcommitment affects double scheduling.

6. Procedure — The paper you studied and what it taught you about existing approaches to double scheduling.

7. Procedure — The M2 lab you did to understand how KVM and the host scheduler interact in practice.

8. Procedure — The kernel source code you read, which files/functions (e.g., CFS scheduler in `kernel/sched/fair.c`, KVM's vCPU handling in `virt/kvm/`), and what you identified.

9. Variables — What you intend to measure: scheduling latency, vCPU wait time, guest process turnaround time. What you vary: number of vCPUs, host load, overcommitment ratio.

10. Statistical analysis — How many times you'll repeat experiments, how you'll report results (means, medians, confidence intervals).

Since you're still in the early phase, elements 6–8 are your main content right now, while 9–10 will come later when you start experimenting. Does that make sense?

the team developped a vm obsevrer "goodkit", you think it will be tested with the VM linux explore ?

Apr 7

That's a reasonable guess. From your past conversations, you've been working with `linux_explore` to build a custom kernel with QEMU/KVM, and you already explored VM observers (like Singularity) in that same environment. So `linux_explore` is your ready-made testbed — you have a host kernel, a guest kernel, and full control over both.

It would make sense for your team to test "goodkit" there because you can instrument the host-side scheduler to observe how vCPUs are being scheduled, and simultaneously observe from inside the guest what the GK is doing with its processes. That's exactly the dual perspective you need to measure double scheduling.

That said, I don't have specific knowledge about "goodkit" itself — is it a kernel module? Does it hook into `ftrace` like Singularity did, or does it work differently (e.g., using KVM tracepoints, perf events)? Knowing how it observes would help me tell you

whether `linux_explore` is sufficient or if you'd need a more complex multi-VM setup.

but the intro should be like 1/2 pages right ?

Apr 7

Yes, roughly 1 to 2 pages is standard for an internship report intro. It should cover:

- The general context (virtualization is everywhere, scheduling matters for performance)
- The specific problem (semantic gap → double scheduling between HK and GK)
- Why it matters (performance degradation, wasted CPU time)
- Your objective (measure and remedy double scheduling)
- A brief roadmap of the report ("Section 2 covers..., Section 3 presents...")

Keep it concise and focused. Don't go into technical details — that's what the background and methodology sections are for. The intro just sets the stage and tells the reader *what* and *why*, not *how*.

but the whole report should be 6 pages the prof said

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For a 6-page report, you need to be much more concise. A rough breakdown:

- **Intro:** half a page to 3/4 page max
- **Background/Related work:** about 1 page (explain virtualization, semantic gap, double scheduling)
- **Methodology:** 1.5 to 2 pages (what you did, how, with what tools)

- **Results / Current progress:** 1 to 1.5 pages (what you found so far, even if preliminary)
- **Conclusion / Future work:** half a page

So your intro should be more like half a page — a short paragraph for context, one for the problem, one sentence for your objective, and a quick roadmap. No filler, every sentence earns its place.

suggest an intro for (respecting the stages in the doc)

Apr 7

Here's a draft intro for your report:

Virtualization has become a fundamental building block in modern computing infrastructures, from cloud platforms to high-performance computing environments. It enables multiple virtual machines to share the same physical hardware by introducing an abstraction layer managed by a hypervisor.

However, this abstraction introduces what is known as the semantic gap: the host kernel and the guest kernel operate with different views of the same resources. A concrete manifestation of this gap is the double scheduling problem. The host kernel schedules vCPUs as ordinary processes, unaware that each vCPU carries the workload of an entire virtual processor. Meanwhile, the guest kernel schedules its own processes on these vCPUs, assuming they are real physical CPUs. This two-layered scheduling leads to inefficiencies such as increased latency and wasted CPU cycles.

The objective of this internship, carried out at [team/lab name], is to measure and mitigate the impact of double scheduling in KVM-based virtualized environments. To this end, we first studied existing literature on the topic, then familiarized ourselves with the virtualization stack through hands-on experimentation, and finally analyzed the relevant kernel

source code to identify potential remediation strategies.

The remainder of this report is organized as follows. Section 2 provides background on virtualization and the double scheduling problem. Section 3 describes our methodology. Section 4 presents our preliminary findings, and Section 5 concludes with future directions.

That's about half a page, which leaves room for the rest. Want me to adjust anything?

precise each stage is what and the plan

Apr 7

Here's the intro again, with each stage labeled:

Stage 1 – General context (broad to narrow): Virtualization has become a fundamental building block in modern computing infrastructures, from cloud platforms to high-performance computing environments. It enables multiple virtual machines to share the same physical hardware by introducing an abstraction layer managed by a hypervisor.

Stage 2 – The specific problem: However, this abstraction introduces what is known as the semantic gap: the host kernel and the guest kernel operate with different views of the same resources. A concrete manifestation of this gap is the double scheduling problem. The host kernel schedules vCPUs as ordinary processes, unaware that each vCPU carries the workload of an entire virtual processor. Meanwhile, the guest kernel schedules its own processes on these vCPUs, assuming they are real physical CPUs. This two-layered scheduling leads to inefficiencies such as increased latency and wasted CPU cycles.

Stage 3 – Objective of the work: The objective of this internship, carried out at [team/lab name], is to measure and

mitigate the impact of double scheduling in KVM-based virtualized environments. To this end, we first studied existing literature on the topic, then familiarized ourselves with the virtualization stack through hands-on experimentation, and finally analyzed the relevant kernel source code to identify potential remediation strategies.

Stage 4 – Roadmap of the report: The remainder of this report is organized as follows. Section 2 provides background on virtualization and the double scheduling problem. Section 3 describes our methodology. Section 4 presents our preliminary findings, and Section 5 concludes with future directions.

So the flow is: **why it matters → what's wrong → what we do about it → how the report is structured**. Want me to refine any part?

pdf

this is the template. I think the intro should be how many pages ? is it written in the doc ?

Apr 7

Let me check your LaTeX template to see if there are any page guidelines.

The template doesn't specify how long the intro should be — it just confirms the total is 6 pages. Neither does the introduction slides document give a specific page count for the intro.

So for a 6-page two-column paper in this IJCAI format (which is quite dense), the intro should be about half a column to one column — roughly half a page. That's enough for the five stages from the slides:

1. **Stage 1** – General context (virtualization, cloud, resource sharing) — 2-3 sentences

2. **Stage 2** – What others have studied (semantic gap, existing scheduling approaches) — 2-3 sentences
3. **Stage 3** – The gap / need for more work (double scheduling still poorly understood/measured) — 1-2 sentences
4. **Stage 4** – Your objective (measure and remedy double scheduling in KVM) — 1-2 sentences
5. **Stage 5** – Value/justification (optional, e.g., improving VM performance benefits cloud providers) — 1 sentence

Then a short roadmap paragraph. That's about half a page in two-column format, leaving 5.5 pages for everything else.

Want me to rewrite the intro following these exact five stages?
instructions

Apr 7

OK so this is a **mid-report**, not the final report, and it's **2 pages max including references** in the two-column IJCAI format.

That's very tight. Here's how I'd split it:

Section 1 – Research problem and objectives (~0.5 column):

This is basically your intro. Stages 1-4 from the slides, but compressed to one short paragraph. Context → problem (double scheduling) → objective.

Section 2 – Background and state of the art (~0.5-0.75 column): Explain virtualization, the semantic gap, how HK schedules vCPUs as processes, what the GK assumes. Mention the paper you read and any existing approaches.

Section 3 (optional) – Current status (~0.5 column): The linux_explore lab, what you learned, the kernel code you've been reading, goodkit. This shows progress.

Section 4 (optional) – Plan (~2-3 sentences): What you'll do next — run experiments, measure overhead, propose a fix.

References (~0.25 column): Keep it to 3-5 key references.

With 2 pages in two-column format, that's about 4 columns total. Every sentence counts — no filler, no "since the dawn of computing" beginnings. Want me to draft something?

I can talk about this ? Introspection has featured prominently in many recent security solutions, such as virtual machine-based intrusion detection, forensic memory analysis, and low-artifact malware analysis. Widespread adoption of these approaches, however, has been hampered by the semantic gap: in order to extract meaningful information about the current state of a virtual machine, detailed knowledge of the guest operating system's inner workings is required. In this paper, we present a novel approach for automatically creating introspection tools for security applications with minimal human effort. By analyzing dynamic traces of small, in-guest programs that compute the desired introspection information, we can produce new programs that retrieve the same information from outside the guest virtual machine. We demonstrate the efficacy of our techniques by automatically generating 17 programs that retrieve security information across 3 different operating systems, and show that their functionality is unaffected by the compromise of the guest system. Our technique allows introspection tools to be effortlessly generated for multiple platforms, and enables the development of rich introspection-based security applications.

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You can mention it, but be careful — this paper is about the semantic gap in the context of **security and introspection**, not scheduling. It's relevant as background to explain what the semantic gap is in general, but it's not directly about your problem (double scheduling).

For a 2-page mid-report, I'd use it like this: one sentence max,

something like "The semantic gap has been widely studied in the context of VM security and introspection [ref], where the host lacks knowledge of guest-level semantics." Then immediately pivot to your specific case: "This same gap also manifests in CPU scheduling, where the host kernel treats vCPUs as ordinary processes, unaware of the guest-level scheduling decisions they carry."

Don't spend more than one sentence on it — your space is too limited to discuss a security paper in detail when your topic is scheduling. If you have papers specifically about double scheduling (e.g., co-scheduling, vCPU scheduling in KVM, or work on scheduler-aware virtualization), those should take priority in your Stage 2.

Do you have any papers that are directly about the double scheduling problem?

what to type on google or google scholar o find smthg ?

Apr 8

Try these queries:

On Google Scholar: "double scheduling virtual machine", "vCPU scheduling KVM", "co-scheduling virtualization", "scheduler-aware virtualization", "host guest scheduling interference"

More specific ones: "CFS scheduler KVM vCPU", "scheduling semantic gap virtualization", "vCPU overcommitment performance", "lock holder preemption KVM"

The term "double scheduling" might not always be used — researchers often call it "two-level scheduling" or "hierarchical scheduling" in virtualized environments. So also try: "two-level scheduling virtual machine", "hierarchical CPU scheduling virtualization".

The "lock holder preemption" one is a well-known consequence of double scheduling where a vCPU holding a spinlock gets

preempted by the host, causing all other guest CPUs to spin-wait — that's a concrete, well-studied symptom you could cite.

Abstract The virtualization technology makes it feasible that multiple guest operating systems run on a single physical machine. It is the virtual machine monitor that dynamically maps the virtual CPU of virtual machines to physical CPUs according to the scheduling strategy. The scheduling strategy in Xen schedules virtual CPUs of a virtual machines asynchronously while guarantees the proportion of the CPU time corresponding to its weight, maximizing the throughput of the system. However, this scheduling strategy may deteriorate the performance when the virtual machine is used to execute the concurrent applications such as parallel programs or multithreaded programs. In this paper, we analyze the CPU scheduling problem in the virtual machine monitor theoretically, and the result is that the asynchronous CPU scheduling strategy will waste considerable physical CPU time when the system workload is the concurrent application. Then, we present a hybrid scheduling framework for the CPU scheduling in the virtual machine monitor. There are two types of virtual machines in the system: the high-throughput type and the concurrent type. The virtual machine can be set as the concurrent type when the majority of its workload is concurrent applications in order to reduce the cost of synchronization. Otherwise, it is set as the high-throughput type as the default. Moreover, we implement the hybrid scheduling framework based on Xen, and we will give a description of our implementation in details. At last, we test the performance of the presented scheduling framework and strategy based on the multi-core platform, and the experiment result indicates that the scheduling framework and strategy is feasible to improve the performance of the virtual machine system.

Apr 8

This is a good find — it's directly about the scheduling problem in virtualization. It focuses on Xen rather than KVM, but the core issue is the same: the VMM schedules vCPUs asynchronously without awareness of what the guest is doing, which hurts concurrent workloads.

You can use this in your Stage 2 to say something like:

"Previous work has shown that asynchronous vCPU scheduling in Xen degrades performance for concurrent guest applications [ref], and proposed hybrid scheduling frameworks to address this."

What's the title and authors so you can cite it properly? And keep looking — try to find at least one paper specifically about KVM/Linux CFS scheduling of vCPUs, since that's your actual platform.

@inproceedings{10.1145/1508293.1508309, author = {Weng, Chuliang and Wang, Zhigang and Li, Minglu and Lu, Xinda}, title = {The hybrid scheduling framework for virtual machine systems}, year = {2009}, isbn = {9781605583754}, publisher = {Association for Computing Machinery}, address = {New York, NY, USA}, url = {<https://doi.org/10.1145/1508293.1508309>}, doi = {10.1145/1508293.1508309}, abstract = {The virtualization technology makes it feasible that multiple guest operating systems run on a single physical machine. It is the virtual machine monitor that dynamically maps the virtual CPU of virtual machines to physical CPUs according to the scheduling strategy. The scheduling strategy in Xen schedules virtual CPUs of a virtual machines asynchronously while guarantees the proportion of the CPU time corresponding to its weight, maximizing the throughput of the system. However, this scheduling strategy may deteriorate the performance when the virtual machine is used to execute the concurrent applications such as parallel programs or multithreaded programs. In this paper, we analyze the CPU scheduling problem in the virtual machine monitor theoretically, and the result is that the

asynchronous CPU scheduling strategy will waste considerable physical CPU time when the system workload is the concurrent application. Then, we present a hybrid scheduling framework for the CPU scheduling in the virtual machine monitor. There are two types of virtual machines in the system: the high-throughput type and the concurrent type. The virtual machine can be set as the concurrent type when the majority of its workload is concurrent applications in order to reduce the cost of synchronization. Otherwise, it is set as the high-throughput type as the default. Moreover, we implement the hybrid scheduling framework based on Xen, and we will give a description of our implementation in details. At last, we test the performance of the presented scheduling framework and strategy based on the multi-core platform, and the experiment result indicates that the scheduling framework and strategy is feasible to improve the performance of the virtual machine system.}, booktitle = {Proceedings of the 2009 ACM SIGPLAN/SIGOPS International Conference on Virtual Execution Environments}, pages = {111–120}, numpages = {10}, keywords = {virtualization, scheduling strategy, hybrid scheduling}, location = {Washington, DC, USA}, series = {VEE '09} }

Apr 8

Good, that's a solid reference — VEE is a reputable venue. Now keep searching for more, ideally one or two about KVM specifically. Try "KVM vCPU scheduling Linux CFS" or "scheduling overhead KVM" on Google Scholar.

too big no ?

Apr 8

The abstract is long yes, but you don't need to summarize the whole paper. In your 2-page report, you'd just use it in one sentence like:

"Weng et al. [2009] showed that asynchronous vCPU scheduling in Xen wastes physical CPU time for concurrent workloads, and proposed a hybrid scheduling framework to mitigate this."

That's it — one sentence, one citation. Save your space for your own work.

Chuliang Weng, Zhigang Wang, Minglu Li, and Xinda Lu. 2009. The hybrid scheduling framework for virtual machine systems. In Proceedings of the 2009 ACM SIGPLAN/SIGOPS international conference on Virtual execution environments (VEE '09). Association for Computing Machinery, New York, NY, USA, 111–120. <https://doi.org/10.1145/1508293.1508309>

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That's the ACM citation format. For your IJCAI template which uses BibTeX, just paste the `@inproceedings` block you had earlier into your `.bib` file and cite it as `\cite{10.1145/1508293.1508309}` (or rename the key to something shorter like `\cite{weng2009hybrid}`).

Keep searching for more papers — you want 3-5 references total. Try to find one about KVM specifically and one about the semantic gap in general.

pdf

what is its result ?

Apr 8

Their main results are:

Theoretical finding: With asynchronous (non-coscheduling) vCPU scheduling, the makespan for concurrent workloads is at least twice as long as with coscheduling, at worst. This is because vCPUs that need to synchronize waste CPU time waiting for each other when they aren't scheduled

simultaneously.

Experimental findings: They tested three workloads on Xen with a 4-vCPU VM:

- **Concurrent workloads (LU, Barnes):** The coscheduling strategy (CON) significantly outperforms Xen's default credit scheduler, especially as contention increases (when Dom0's weight is higher relative to Dom1). For example, LU completion time roughly doubles under the credit scheduler compared to coscheduling when contention is high.
- **Throughput workload (web server):** The default credit scheduler (HIT) performs better than coscheduling, because independent requests don't need synchronization.

Bottom line: No single scheduling strategy fits all workloads. Their hybrid framework, which lets you tag a VM as "concurrent" or "high-throughput" and applies the appropriate strategy, consistently picks the better option. This is directly relevant to your work — it confirms that the two-level scheduling problem causes real, measurable performance loss, and that the solution requires awareness of guest-level workload semantics.

explain and tell me which of these docs potentially talks about it

Apr 8

4-related-work.pdf is the one you want. It covers how to write about existing work, cite other papers, and position your contribution relative to what's been done before.

2-article-structure-and-resear...pdf might also help — it likely covers the overall structure of a research paper, including where background/state of the art fits.

For your Section 2, you need to:

- Explain the key concepts (virtualization, semantic gap, double

scheduling) — that's background

- Summarize what others have done about it (Weng et al., the introspection paper, etc.) — that's state of the art

The "4-related-work.pdf" doc will teach you *how* to write about other people's work properly: how to cite, how to compare, how to show what's missing (which justifies your internship's contribution). Want me to look at it if you upload it?

State of art is stage 2 i think

Apr 8

Yes exactly. In the intro, Stage 2 is a brief mention of existing work (1-2 sentences). But in your mid-report, you also have a dedicated **Section 2** for background and state of the art, where you go into more detail.

So the difference is:

In the intro (Stage 2): "Previous work has addressed vCPU scheduling inefficiencies in Xen [Weng et al., 2009] and studied the semantic gap in the context of VM introspection [ref]. However, ..." → just a quick nod, then move to Stage 3 (the gap).

In Section 2 (dedicated background): You actually explain the concepts and the papers in more depth — what virtualization is, how the HK/GK scheduling works, what Weng et al. found, what's still missing (especially for KVM, since their work was on Xen).

So Stage 2 in the intro is like a teaser, and Section 2 is the full picture. Don't repeat the same content twice — keep the intro short and let Section 2 do the heavy lifting.

So technically I have to write the intro

Apr 8

Yes, but it's not labeled "Introduction" in the traditional sense. Section 1 is your intro — it covers the research problem and objectives, which maps directly to the five stages:

- Stage 1: general context (virtualization)
- Stage 2: what others studied (brief)
- Stage 3: what's missing (gap)
- Stage 4: your objective
- Stage 5: justification (optional)

Then Section 2 is where you expand on background and state of the art in more detail.

So for 2 pages, think of it as: Section 1 (problem + objectives) is your intro compressed into maybe 1/3 of a column, and Section 2 (background + state of the art) gets more space since it's where the substance is. The optional sections 3 and 4 would only get a few sentences each if you include them.

Want to start drafting Section 1?

i want the introductions to be around 1 page (2 columns in this template) suggest smthg

Apr 8

Here's a draft for a 1-page (two-column) introduction:

1. Research Problem and Objectives

Virtualization has become a cornerstone of modern computing infrastructures, enabling multiple virtual machines (VMs) to share the same physical hardware. In such environments, a hypervisor — or virtual machine monitor (VMM) — mediates access to physical resources by exposing virtual CPUs (vCPUs) to each guest operating system. This architecture is widely deployed in cloud platforms, data centers, and high-performance computing environments, where resource

consolidation and isolation are critical requirements.

However, virtualization introduces a fundamental challenge known as the *semantic gap*: the host kernel (HK) and the guest kernel (GK) operate with fundamentally different views of the same resources. The semantic gap has been studied extensively in the context of VM security and introspection, where extracting meaningful information about a guest's state requires detailed knowledge of its internal structures [ref introspection paper]. This same gap also manifests in CPU scheduling. From the host's perspective, each vCPU is simply a process — or more precisely, a thread — that competes for physical CPU time alongside all other processes on the host. The host scheduler, such as the Linux Completely Fair Scheduler (CFS), has no awareness that these threads represent virtual processors running an entire operating system. Meanwhile, the guest kernel treats its vCPUs as real, physical CPUs and schedules its own processes on them accordingly.

This discrepancy leads to what is known as *double scheduling*: the host kernel first decides when a vCPU may run on a physical CPU, and then the guest kernel schedules a process onto that vCPU, unaware of the underlying scheduling constraints. As a result, scheduling decisions are made at two independent levels without coordination, leading to inefficiencies such as increased latency, wasted CPU cycles, and degraded application performance. Previous work has demonstrated the severity of this problem in the Xen hypervisor, where asynchronous vCPU scheduling was shown to waste considerable physical CPU time for concurrent workloads, with makespan at least doubling compared to a coscheduling approach [Weng et al., 2009]. However, most existing studies have focused on Xen-based environments, and the double scheduling problem remains insufficiently studied in the context of KVM, where the host Linux kernel directly schedules vCPUs using CFS.

The objective of this internship, carried out at [team/lab name,

e.g., LIG, Inria Grenoble], is to measure the impact of double scheduling in KVM-based virtualized environments and to explore strategies to mitigate it. To this end, we leverage a custom kernel experimentation platform and VM observation tools developed by the team to analyze scheduling behavior from both the host and guest perspectives.

The remainder of this report is organized as follows. Section 2 provides background on virtualization and the scheduling mechanisms involved. Section 3 describes our current progress and future plans.

That should fill roughly one page in two-column IJCAI format. It covers all five stages and naturally weaves in your two references. Want me to adjust anything?

is this sentence correct ? It allows for the creation of virtual environments. Before virtualization, we could only run one operating system on a physical server, leading to a considerable waste of resources. Now, it is possible to run several operating systems by emulating the physical hardware pf a machine. However, virtual machines still present several drawbacks.

Apr 8

A few fixes needed:

- "pf" → "of" (typo)
- "It allows for the creation of virtual environments" is a bit vague as an opening — what is "it"? Make sure the previous sentence introduces virtualization.
- "Before virtualization, we could only run one operating system" — avoid "we" in academic writing, use passive or impersonal form
- "leading to a considerable waste of resources" — fine
- "Now, it is possible to run several operating systems by

emulating the physical hardware of a machine" — this is a bit imprecise. Virtualization doesn't just "emulate" hardware in all cases (KVM uses hardware-assisted virtualization). Better to say something like "by abstracting the underlying hardware."

Here's a corrected version:

"Virtualization allows for the creation of virtual environments. Before its introduction, only one operating system could run on a given physical server, leading to a considerable waste of resources. Today, it is possible to run several operating systems simultaneously on a single machine by abstracting the underlying physical hardware. However, virtual machines still present several drawbacks."

That said, for a 2-page mid-report this is quite generic and takes up precious space. You might want to compress these four sentences into one or two and get to the problem faster.

describe the pb according to the doc of the internship

Apr 8

Based on what you shared earlier in this conversation, the problem from your internship description is:

"In the context of virtualization, the semantic gap manifests through the double scheduling of processes. The host kernel (HK) schedules vCPUs, which are then used by the guest kernel (GK) to schedule processes within the VM. The HK perceives vCPUs as ordinary processes, while the GK treats them as actual physical CPUs. This leads to double scheduling: the HK first schedules a vCPU onto a physical CPU, and then the GK schedules a guest process onto that vCPU, without any coordination between the two layers. The goal of this internship is to measure and remedy this double scheduling."

For your mid-report, you could write it more academically like this:

"One such drawback is the *double scheduling* problem, which arises from the semantic gap between the host and the guest. The host kernel schedules each vCPU as an ordinary process, using its standard scheduler (e.g., CFS on Linux), without any knowledge of what runs inside the VM. Meanwhile, the guest kernel assumes its vCPUs are real physical CPUs and schedules its own processes on them independently. These two scheduling layers operate without coordination, resulting in inefficiencies such as increased latency and wasted CPU time. The objective of this internship is to quantify the overhead introduced by this double scheduling in KVM-based environments and to propose strategies to mitigate it."

This connects directly to your previous paragraph about virtualization drawbacks. Want me to integrate everything into a full draft?

reexplain the stages of the intro, and suggest a full intro (comment the name of each stage)

Apr 10

Here are the five stages, then the full draft:

Stage 1 – General context: Broad, accepted facts about your field to set the scene (universe → galaxy → star)

Stage 2 – What others have studied: Specific statements about existing work on the problem

Stage 3 – The gap: What's still missing, why more work is needed

Stage 4 – Your objective: The precise purpose of your study

Stage 5 – Value/justification (optional): Why this work matters

Full intro draft:

% --- STAGE 1: General context (universe → galaxy → star) ---

Virtualization has become a fundamental technology in modern computing, enabling multiple operating systems to run simultaneously on a single physical machine. By abstracting the underlying hardware, a hypervisor — or virtual machine monitor (VMM) — creates isolated virtual machines (VMs), each equipped with virtual CPUs (vCPUs), virtual memory, and virtual devices. This approach is now widely deployed in cloud computing, data centers, and high-performance computing environments, where efficient resource sharing and workload consolidation are essential.

However, this level of abstraction introduces a fundamental challenge known as the semantic gap: the host and the guest operate with different views of the same resources. In the case of CPU scheduling, the host kernel (HK) treats each vCPU as an ordinary process and schedules it on physical CPUs using its standard scheduler, such as the Linux Completely Fair Scheduler (CFS). Meanwhile, the guest kernel (GK) assumes its vCPUs are real physical CPUs and schedules its own processes on them independently. Since these two scheduling layers operate without coordination, a phenomenon known as double scheduling arises: the HK first decides when a vCPU may run, and then the GK assigns a guest process to that vCPU, unaware of the host-level constraints. This leads to inefficiencies such as increased latency, wasted CPU cycles, and degraded application performance.

% --- STAGE 2: What others have studied ---

Previous work has studied this problem primarily in the Xen hypervisor. Weng et al. \cite{weng2009hybrid} demonstrated that asynchronous vCPU scheduling wastes considerable physical CPU time when running concurrent workloads, with execution time at least doubling compared to a coscheduling approach. They proposed a hybrid scheduling framework that classifies VMs as either high-throughput or concurrent and applies different scheduling strategies accordingly. The

semantic gap has also been studied in the context of VM security and introspection, where extracting guest-level information from outside the VM requires detailed knowledge of the guest's internal structures \cite{introspection_ref}.

% --- STAGE 3: The gap / need for more investigation ---

However, most existing studies have focused on Xen-based environments with custom VMM schedulers. The double scheduling problem remains insufficiently studied in the context of KVM, where the host Linux kernel directly schedules vCPUs as standard threads through CFS, with no built-in awareness of guest-level scheduling decisions.

% --- STAGE 4: Objective of the study ---

The objective of this internship, carried out at [team/lab name], is to measure the overhead introduced by double scheduling in KVM-based virtualized environments and to explore strategies to mitigate it. To this end, we leverage a custom kernel experimentation platform and VM observation tools developed by the team to analyze scheduling behavior from both the host and guest perspectives.

% --- STAGE 5: Value / justification (optional) ---

Understanding and addressing double scheduling in KVM is particularly relevant given the widespread adoption of KVM in production cloud environments, where even small scheduling inefficiencies can compound across thousands of virtual machines.

% --- ROADMAP ---

The remainder of this report is organized as follows. Section 2 provides background on virtualization and the scheduling mechanisms involved. Section 3 describes our current progress, and Section 4 outlines our plan for the remainder of the internship.

This should fill roughly one page in two-column format. Want

me to adjust any part?

in the state of art I put a paragraph on each of the 2/3 papers I read till now in my internship right ?

Apr 10

Yes, exactly. For each paper, write a short paragraph (3-5 sentences) covering:

- What problem they addressed
- What they proposed/did
- What their main result was
- What limitation remains (this connects back to why your work is needed)

For your 2-page mid-report, keep it tight — maybe 2-3 papers max with one paragraph each. For example:

Paper 1 (Weng et al.): vCPU scheduling in Xen, hybrid framework, coscheduling vs async → but only Xen, not KVM

Paper 2 (introspection paper): semantic gap in VM security context → shows the gap is a general problem, not just scheduling

Paper 3 (if you find one on KVM): anything about KVM scheduling overhead → directly relevant to your platform

Then close the state of the art with a sentence like: "To the best of our knowledge, the double scheduling overhead in KVM using the Linux CFS scheduler has not been systematically measured, which motivates the present work."

That sentence bridges your state of the art back to your objective. Does that structure work for you?

pdf

Resume this

Apr 10

This paper by Zhao et al. (Middleware 2017) addresses the same core problem as your internship — the semantic gap in vCPU scheduling — but focuses on a specific consequence: **lock-holder preemption (LHP)** and **lock-waiter preemption (LWP)**.

The problem: When the hypervisor preempts a vCPU that is holding a lock (LHP) or waiting for a lock (LWP), all other threads depending on that lock stall until the vCPU is rescheduled. This causes significant performance degradation for parallel workloads — up to 3.5x slowdown.

Key insight: Previous work tried to fix this from the hypervisor side (coscheduling, heuristics). This paper identifies a *reverse* semantic gap: the guest OS is also unaware of hypervisor scheduling events. The guest's own load balancer fails because it sees preempted threads as "running" and doesn't trigger migration.

Their solution (IRS): Inspired by scheduler activations, they notify the guest OS *before* a vCPU is preempted. The guest then migrates the critical thread to another running vCPU. This requires minimal changes — about 30 lines in Xen and 130 lines in the Linux guest.

Results: Up to 42% improvement for blocking workloads (PARSEC), 43% for spinning workloads (NPB), and 46% latency reduction for server workloads. IRS works best when only some vCPUs are under interference; when all vCPUs are contended, gains diminish.

Limitation for your work: This is Xen-based, not KVM. And it addresses a *consequence* of double scheduling (LHP/LWP) rather than measuring the double scheduling overhead itself. Still, it's an excellent reference — it directly shows how the semantic gap hurts scheduling and proposes a guest-hypervisor coordination approach.

last sentence of my introduction During this internship, the reminder of this study aims to find a new efficient way to reduce the semantic gap between the host kernel and the guest kernel. To achieve that, we have to understand what information and data should the vm observer, \textbf{GOODKIT} in our case, send to the host kernel.

Apr 10

A few corrections:

- "the reminder" → "the remainder" (typo)
- "what information and data should the vm observer" → "what information and data the VM observer should" (indirect question word order)
- Don't mix the objective with methodology details in the intro — mentioning GOODKIT by name feels too specific here

Suggested revision:

"The objective of this internship is to find an efficient approach to reducing the semantic gap between the host kernel and the guest kernel in the context of CPU scheduling. In particular, we investigate what information a VM observer should communicate to the host kernel to enable scheduling-aware decisions. To this end, we rely on GOODKIT, a VM observation tool developed by the team."

This separates the objective (what you want to achieve) from the method (how you'll do it), which is cleaner. You can then detail GOODKIT in the methodology or background section.

before i start talking about articles in this part, should i put uinder this title \section{Background information and the state of the art} a little paragraph as an "intro " for this part ? and should I talk about working on vbirtualization lab from M2 ?

Apr 10

Yes to the intro paragraph, no to the M2 lab.

Intro paragraph: Write 2-3 sentences introducing what this section covers. Something like: "This section provides the necessary background on virtualization and CPU scheduling in virtual environments, followed by a review of existing approaches to the double scheduling problem."

M2 lab: Don't mention it here. The state of the art is about *other people's work*, not your learning process. The M2 lab is part of your methodology (how you got up to speed), not background. If you include Section 3 (current status), you can briefly mention it there as part of your experimental setup familiarization.

For this section, structure it as:

1. Short intro paragraph (what this section covers)
2. Brief background on virtualization and the two-level scheduling mechanism (3-4 sentences explaining HK/GK, vCPUs, CFS — the concepts the reader needs)
3. One paragraph per paper (Weng et al., Zhao et al., possibly the introspection paper)
4. A closing sentence highlighting what's still missing (the gap that justifies your work)

Keep it tight — you have maybe 0.5 to 0.75 column for this entire section.

More developped

Apr 10

Here's a more developed draft for Section 2:

\section{Background Information and State of the Art}

Intro paragraph: "In this section, we first introduce the key concepts of virtualization and the scheduling mechanisms

involved in the double scheduling problem. We then review existing approaches that have been proposed to address scheduling inefficiencies in virtualized environments."

Background subsection (no need for a formal subsection heading given space constraints, just a paragraph):

"In a virtualized system, a hypervisor or virtual machine monitor (VMM) sits between the hardware and the guest operating systems, abstracting physical resources into virtual ones. Each virtual machine is assigned one or more virtual CPUs (vCPUs), on which the guest kernel schedules its processes. On the host side, these vCPUs are treated as ordinary threads and scheduled onto physical CPUs by the host kernel's scheduler. In the case of KVM on Linux, the host uses the Completely Fair Scheduler (CFS), which has no knowledge of what runs inside the VM. This creates two independent scheduling layers that operate without coordination. The guest kernel makes scheduling decisions assuming its vCPUs are real physical CPUs that are always available, while the host kernel may preempt a vCPU at any time, unaware that it may be running a critical thread or holding a lock inside the guest. This disconnect is a manifestation of the semantic gap between host and guest, and is the root cause of the double scheduling problem."

Paper 1 — Weng et al. (VEE 2009):

"Weng et al. \cite{weng2009hybrid} studied the impact of asynchronous vCPU scheduling in the Xen hypervisor. They demonstrated theoretically and experimentally that when a VM runs concurrent workloads such as multithreaded or parallel programs, the default proportional share scheduling strategy wastes considerable physical CPU time due to synchronization delays between vCPUs. In the worst case, the execution time of concurrent workloads at least doubles compared to a coscheduling approach, where all vCPUs of a VM are scheduled simultaneously. To address this, they proposed a hybrid

scheduling framework that classifies VMs into two types — high-throughput and concurrent — and applies different scheduling strategies accordingly. Their experiments on Xen showed significant performance improvements for concurrent workloads. However, their approach requires modifications to the VMM scheduler and was only evaluated on Xen, leaving the problem open for KVM-based environments."

Paper 2 — Zhao et al. (Middleware 2017):

"Zhao et al. \cite{zhao2017irs} investigated two well-known consequences of the double scheduling problem: lock-holder preemption (LHP) and lock-waiter preemption (LWP). LHP occurs when the hypervisor preempts a vCPU whose thread is holding a lock, causing all other threads waiting for that lock to stall. LWP is a similar issue where a thread waiting for a lock is preempted. The authors identified a reverse semantic gap — the guest OS is unaware of scheduling events at the hypervisor — and proposed Interference-Resilient Scheduling (IRS), a mechanism inspired by scheduler activations. IRS notifies the guest OS before a vCPU is preempted, allowing the guest to migrate critical threads to other running vCPUs. Their prototype, implemented in Xen and Linux, achieved up to 42% performance improvement for parallel workloads with blocking synchronization. While IRS demonstrates the benefits of guest-hypervisor coordination, it focuses on mitigating the consequences of vCPU preemption rather than measuring the overall overhead of double scheduling, and it was not evaluated in a KVM environment."

Closing paragraph (the gap):

"While these studies highlight the severity of scheduling inefficiencies in virtualized environments, they share two common limitations. First, they were conducted exclusively on Xen-based platforms, where the hypervisor has direct control over vCPU scheduling. In KVM, the situation differs significantly because vCPUs are scheduled as regular Linux threads by CFS,

making the problem both harder to detect and harder to address. Second, existing work has focused on proposing solutions without first systematically measuring the overhead introduced by double scheduling. This motivates the present work, which aims to quantify this overhead in KVM and to explore how a VM observer can bridge the semantic gap by communicating guest-level scheduling information to the host kernel."

This should take about 0.75 to 1 column in the two-column format. Want me to adjust the length or any specific paragraph?

How to write an abstract

Apr 12

An abstract is a miniature version of your entire paper. For your 2-page mid-report, it should be about 5-8 sentences (roughly 100-150 words). It follows the same flow as your paper, compressed:

Sentence 1 — Context: What is the general area?

"Virtualization enables multiple operating systems to share the same physical hardware through virtual machines."

Sentence 2 — Problem: What specific issue are you addressing? "However, the semantic gap between the host and guest kernels leads to a double scheduling problem, where vCPUs are scheduled by the host as ordinary processes while the guest independently schedules its own processes on them."

Sentence 3 — What others have done (optional, brief):

"Previous work has addressed this issue in Xen-based environments, but the problem remains insufficiently studied in the context of KVM."

Sentence 4 — Your objective: "This internship aims to measure the overhead introduced by double scheduling in KVM

and to explore strategies to mitigate it."

Sentence 5 — Your approach: "We leverage GOODKIT, a VM observation tool, to analyze scheduling behavior from both the host and guest perspectives."

Sentence 6 — Current status / preliminary results (if any):

"Preliminary analysis of the Linux CFS scheduler and KVM source code has identified key areas where guest-level information could improve host scheduling decisions."

Key rules: write it last (after the whole paper is done), use past tense for what you did, present tense for general facts, and don't include references or details not in the paper.

the goodkit paper, done by the research team i'm intern with, the paper is not officially published but it was accepted. so i don't have the full paper's title to write a reference. what to do ? do i go to the lab ask the superv or is there a format for this

8:41 AM

Ask your supervisor. This is the right thing to do for two reasons:

First, they'll give you the exact title, authors, and venue so you can cite it properly. BibTeX has a format for accepted but not yet published papers:

```
@article{goodkit2025,  
  author = {Author1 and Author2},  
  title = {Title of the Paper},  
  journal = {Name of the Venue},  
  year = {2025},  
  note = {To appear}  
}
```

The key is the `note = {To appear}` field, which signals that the paper has been accepted but not yet published. Some people also use `note = {Accepted, to be published}`.

Second, and more importantly, you should ask because some supervisors prefer that you **don't cite** unpublished work in a student report, especially if the paper is still under embargo or they don't want the title public yet. Others will be happy for you to reference it. It's a quick question — just ask something like "I'd like to cite the GOODKIT paper in my mid-report. Could you give me the full reference and confirm it's okay to cite it as accepted?"

This also shows your supervisor you're making progress on the report, which is always a good signal.

he already told me i can speak about it

8:41 AM

Then just ask him for the exact title, author list, and venue name so you can write the BibTeX entry properly. A quick message like:

"Could you send me the full reference for the GOODKIT paper (title, authors, venue) so I can cite it correctly in my mid-report?"

That's all you need. Once you have it, use:

```
@article{goodkit2025,  
  author = {...},  
  title = {...},  
  journal = {...},  
  year = {2025},  
  note = {To appear}  
}
```

And in your state of the art, you'd write a paragraph about it just like the other papers, something along the lines of:

"[Authors] \cite{goodkit2025} developed GOODKIT, a VM observation tool that [does what]. This tool enables [what capability], and constitutes the foundation of the present work."

You can fill in the details once you have the paper reference. Go

ask him — it's a 30-second question.

found it on the research's team site. how to write it in latex ? %
% The file named.bst is a bibliography style file for BibTeX 0.99c
\bibliographystyle{named} \bibliography{ijcai11} Chuliang
Weng, Zhigang Wang, Minglu Li, and Xinda Lu. 2009. The hybrid
scheduling framework for virtual machine systems. In
Proceedings of the 2009 ACM SIGPLAN/SIGOPS international
conference on Virtual execution environments (VEE '09).
Association for Computing Machinery, New York, NY, USA, 111–
120. <https://doi.org/10.1145/1508293.1508309> Dufy Teguia,
Louis Duval, Teo Pisenti, Kahina Lazri, Daniel Hagimont,
Thomas Pasquier, Renaud Lachaize, Alain Tchana. Inside Out: A
Paradigm Shift In Live VM Introspection, OSDI 2026.

9:25 AM

In your `ijcai11.bib` file, add these two entries:

`bibtex`

```
@inproceedings{weng2009hybrid,  
  author = {Chuliang Weng and Zhigang Wang and  
Minglu Li and Xinda Lu},  
  title = {The Hybrid Scheduling Framework for  
Virtual Machine Systems},  
  booktitle = {Proceedings of the 2009 ACM  
SIGPLAN/SIGOPS International Conference on  
Virtual Execution Environments (VEE)},  
  year = {2009},  
  pages = {111--120},  
  doi = {10.1145/1508293.1508309}  
}  
  
@inproceedings{goodkit2026,  
  author = {Dufy Teguia and Louis Duval and Teo  
Pisenti and Kahina Lazri and Daniel Hagimont and
```

```
Thomas Pasquier and Renaud Lachaize and Alain  
Tchana},  
  title = {Inside Out: A Paradigm Shift In Live  
VM Introspection},  
  booktitle = {Proceedings of the 16th USENIX  
Symposium on Operating Systems Design and  
Implementation (OSDI)},  
  year = {2026},  
  note = {To appear}  
}
```

Then in your LaTeX text, cite them as

`\cite{weng2009hybrid}` and `\cite{goodkit2026}`.

Don't forget to also add the Zhao et al. paper if you use it:

bibtex

```
@inproceedings{zhao2017irs,  
  author = {Yong Zhao and Kun Suo and Luwei Cheng  
and Jia Rao},  
  title = {Scheduler Activations for  
Interference-Resilient SMP Virtual Machine  
Scheduling},  
  booktitle = {Proceedings of the 18th ACM/IFIP/  
USENIX International Middleware Conference  
(Middleware)},  
  year = {2017},  
  pages = {65--78},  
  doi = {10.1145/3135974.3135975}  
}
```

OSDI is a top venue by the way — the fact that your team got a paper there is impressive. That's worth mentioning to your prof.

preferably the same format as the one i gave you

9:27 AM

Here are all three in the same format:

bibtex

```
@inproceedings{10.1145/1508293.1508309,  
  author = {Weng, Chuliang and Wang, Zhigang and  
Li, Minglu and Lu, Xinda},  
  title = {The hybrid scheduling framework for  
virtual machine systems},  
  year = {2009},  
  isbn = {9781605583754},  
  publisher = {Association for Computing  
Machinery},  
  address = {New York, NY, USA},  
  url = {https://  
doi.org/10.1145/1508293.1508309},  
  doi = {10.1145/1508293.1508309},  
  booktitle = {Proceedings of the 2009 ACM  
SIGPLAN/SIGOPS International Conference on  
Virtual Execution Environments},  
  pages = {111--120},  
  numpages = {10},  
  keywords = {virtualization, scheduling  
strategy, hybrid scheduling},  
  location = {Washington, DC, USA},  
  series = {VEE '09}  
}  
  
@inproceedings{10.1145/3135974.3135975,  
  author = {Zhao, Yong and Suo, Kun and Cheng,  
Luwei and Rao, Jia},  
  title = {Scheduler activations for  
interference-resilient SMP virtual machine  
scheduling},  
  year = {2017},
```

```

    publisher = {Association for Computing
Machinery},
    address = {New York, NY, USA},
    url = {https://
doi.org/10.1145/3135974.3135975},
    doi = {10.1145/3135974.3135975},
    booktitle = {Proceedings of the 18th ACM/IFIP/
USENIX International Middleware Conference},
    pages = {65--78},
    numpages = {14},
    keywords = {virtualization, cloud computing,
multi-tenancy, semantic gaps},
    location = {Las Vegas, NV, USA},
    series = {Middleware '17}
}

```

```

@inproceedings{teguia2026insideout,
    author = {Tegua, Dufy and Duval, Louis and
Pisenti, Teo and Lazri, Kahina and Hagimont,
Daniel and Pasquier, Thomas and Lachaize, Renaud
and Tchan, Alain},
    title = {Inside Out: A paradigm shift in live
VM introspection},
    year = {2026},
    booktitle = {Proceedings of the 16th USENIX
Symposium on Operating Systems Design and
Implementation},
    series = {OSDI '26},
    note = {To appear}
}

```

Then cite them as \cite{10.1145/1508293.1508309},
\cite{10.1145/3135974.3135975}, and
\cite{teguia2026insideout}.

smthg like this ?

9:27 AM

No, your reference format will be different. Your template uses `\bibliographystyle{named}` which is the IJCAI style — it produces references like:

```
[Weng et al., 2009] Chuliang Weng, Zhigang Wang,
Minglu Li,
and Xinda Lu. The hybrid scheduling framework for
virtual
machine systems. In Proceedings of VEE, 2009.
```

The format in your screenshot uses a different bibliography style (looks like it comes from a different template). You don't control the format — BibTeX handles it automatically based on the `named.bst` file.

All you need to do is:

1. Put the `@inproceedings` entries in your `ijcai11.bib` file
2. Use `\cite{...}` in your text
3. Compile with: `pdflatex main`, then `bibtex main`, then `pdflatex main` twice

BibTeX will format everything automatically to match the IJCAI style. Just make sure your `.bib` file has all the fields filled in properly and let the style file do the rest.

show me

9:28 AM

With the `named.bst` style, your references will look something like this:

References

```
[Weng et al., 2009] Chuliang Weng, Zhigang Wang,
```

Minglu Li,
and Xinda Lu. The hybrid scheduling framework
for virtual
machine systems. In Proceedings of the 2009 ACM
SIGPLAN/SIGOPS
International Conference on Virtual Execution
Environments,
pages 111–120, 2009.

[Zhao et al., 2017] Yong Zhao, Kun Suo, Luwei
Cheng, and
Jia Rao. Scheduler activations for
interference-resilient
SMP virtual machine scheduling. In Proceedings
of the 18th
ACM/IFIP/USENIX International Middleware
Conference,
pages 65–78, 2017.

[Tegua et al., 2026] Dufy Tegua, Louis Duval,
Teo Pisenti,
Kahina Lazri, Daniel Hagimont, Thomas Pasquier,
Renaud
Lachaize, and Alain Tchana. Inside Out: A
paradigm shift
in live VM introspection. In Proceedings of the
16th USENIX
Symposium on Operating Systems Design and
Implementation,
2026. To appear.

And in the text, when you write

`\cite{10.1145/1508293.1508309}`, it will appear as [Weng
et al., 2009]. You can also use

`\shortcite{10.1145/1508293.1508309}` which gives just
[2009] when you already mentioned the authors by name, like:

"Weng et al. \shortcite{10.1145/1508293.1508309} showed that..."

do it with this Dufy Tegua, Louis Duval, Teo Piseni, Kahina Lazri, Daniel Hagimont, Thomas Pasquier, Renaud Lachaize, Alain Tchan. Inside Out: A Paradigm Shift In Live VM Introspection, OSDI 2026.

9:30 AM

In the named .bst style it will appear as:

```
[Tegua et al., 2026] Dufy Tegua, Louis Duval,
Teo Piseni,
    Kahina Lazri, Daniel Hagimont, Thomas Pasquier,
Renaud
    Lachaize, and Alain Tchan. Inside Out: A
paradigm shift
    in live VM introspection. In Proceedings of the
16th USENIX
    Symposium on Operating Systems Design and
Implementation
    (OSDI), 2026. To appear.
```

And in your text you'd write either:

- Tegua et al.~\shortcite{tegua2026insideout} developed GOODKIT... → renders as: "Tegua et al. [2026] developed GOODKIT..."
- \cite{tegua2026insideout} → renders as: "[Tegua et al., 2026]"

just write it %% The file named.bst is a bibliography style file for BibTeX 0.99c \bibliographystyle{named} \bibliography{ijcai11} Chuliang Weng, Zhigang Wang, Minglu Li, and Xinda Lu. 2009. The hybrid scheduling framework for virtual machine systems. In Proceedings of the 2009 ACM SIGPLAN/SIGOPS international conference on Virtual execution environments (VEE '09).

Association for Computing Machinery, New York, NY, USA, 111–120. <https://doi.org/10.1145/1508293.1508309> Dufy Teguia, Louis Duval, Teo Pisenti, Kahina Lazri, Daniel Hagimont, Thomas Pasquier, Renaud Lachaize, Alain Tchana. Inside Out: A Paradigm Shift In Live VM Introspection, OSDI 2026.

\cite{teguia2026insideout} \end{document}

9:32 AM

```
%% The file named.bst is a bibliography style
file for BibTeX 0.99c
\bibliographystyle{named}
\bibliography{ijcai11}
```

Then in your ijcai11.bib file:

bibtex

```
@inproceedings{10.1145/1508293.1508309,
  author = {Weng, Chuliang and Wang, Zhigang and
Li, Minglu and Lu, Xinda},
  title = {The hybrid scheduling framework for
virtual machine systems},
  year = {2009},
  isbn = {9781605583754},
  publisher = {Association for Computing
Machinery},
  address = {New York, NY, USA},
  url = {https://
doi.org/10.1145/1508293.1508309},
  doi = {10.1145/1508293.1508309},
  booktitle = {Proceedings of the 2009 ACM
SIGPLAN/SIGOPS International Conference on
Virtual Execution Environments},
  pages = {111--120},
  series = {VEE '09}
}
```

```
@inproceedings{10.1145/3135974.3135975,  
  author = {Zhao, Yong and Suo, Kun and Cheng,  
Luwei and Rao, Jia},  
  title = {Scheduler activations for  
interference-resilient SMP virtual machine  
scheduling},  
  year = {2017},  
  publisher = {Association for Computing  
Machinery},  
  address = {New York, NY, USA},  
  url = {https://  
doi.org/10.1145/3135974.3135975},  
  doi = {10.1145/3135974.3135975},  
  booktitle = {Proceedings of the 18th ACM/IFIP/  
USENIX International Middleware Conference},  
  pages = {65--78},  
  series = {Middleware '17}  
}
```

```
@inproceedings{teguia2026insideout,  
  author = {Tegua, Dufy and Duval, Louis and  
Pisenti, Teo and Lazri, Kahina and Hagimont,  
Daniel and Pasquier, Thomas and Lachaize, Renaud  
and Tchana, Alain},  
  title = {Inside Out: A Paradigm Shift In Live  
VM Introspection},  
  year = {2026},  
  booktitle = {Proceedings of the 16th USENIX  
Symposium on Operating Systems Design and  
Implementation},  
  series = {OSDI '26},  
  note = {To appear}  
}
```

Then in your text just use

```
\cite{10.1145/1508293.1508309},  
\cite{10.1145/3135974.3135975}, and  
\cite{teguia2026insideout}.
```

like this ? \cite{teguia2026insideout} Dufy Tegua, Louis Duval, Teo Pisenti, Kahina Lazri, Daniel Hagimont, Thomas Pasquier, Renaud Lachaize, Alain Tchana. Inside Out: A Paradigm Shift In Live VM Introspection, OSDI 2026.

9:34 AM

No, you don't write the reference text manually in your .tex file. BibTeX generates it automatically. Here's what goes where:

In main.tex (at the end, before \end{document}):

latex

```
\bibliographystyle{named}  
\bibliography{ijcai11}  
  
\end{document}
```

In ijcai11.bib (a separate file):

bibtex

```
@inproceedings{teguia2026insideout,  
  author = {Tegua, Dufy and Duval, Louis and  
Pisenti, Teo and Lazri, Kahina and Hagimont,  
Daniel and Pasquier, Thomas and Lachaize, Renaud  
and Tchana, Alain},  
  title = {Inside Out: A Paradigm Shift In Live  
VM Introspection},  
  year = {2026},  
  booktitle = {Proceedings of the 16th USENIX  
Symposium on Operating Systems Design and  
Implementation},  
  series = {OSDI '26},
```

```
note = {To appear}
}
```

In your text (wherever you want to cite it):

latex

```
Tegua et al.\shortcite{tegua2026insideout}
developed GOODKIT...
```

That's it. You never write the full reference by hand. BibTeX reads the .bib file and formats it automatically into the References section. Just compile with: `pdflatex main → bibtex main → pdflatex main → pdflatex main`.

I create it ?

11:10 AM

Yes, create a file called `ijcai11.bib` in the same folder as your `main.tex`. It's just a plain text file. Put all your `@inproceedings{...}` entries in it, nothing else.

put my articles under this format : `@article{ACallFor, author = {Sara McAllister and Fiodar Kazhamiaka and Daniel S. Berger and Rodrigo Fonseca and Kali Frost and Aaron Ogus and Maneesh Sah and Ricardo Bianchini and George Amvrosiadis and Nathan Beckmann and Gregory R. Ganger}, title = {A Call for Research on Storage Emissions}, journal = {ACM SIGENERGY Energy Informatics Review}, year = {2024} }` `@article{straid, author = {Shucheng Wang and Qiang Cao and Hong Jiang and Ziyi Lu and Jie Yao and Yuxing Chen and Anqun Pan}, title = {Explorations and Exploitation for Parity-based RAIDs with Ultra-fast SSDs}, journal = {ACM Transactions on Storage}, year = {2024} }` `@inproceedings{scalaraid, author = {Shushu Yi and Yanning Yang and Yunxiao Tang and Zixuan Zhou and Junzhe Li and Chen Yue and Myoungsoo Jung and Jie Zhang}, title = {ScalaRAID: Optimizing Linux Software RAID System for Next-Generation Storage}, booktitle = {Proceedings of the 14th ACM`

Workshop on Hot Topics in Storage and File Systems
(HotStorage '22)}, year = {2022} } @article{flash, author =
{Zhiyue Li and Guangyan Zhang and Yang Wang}, title = {Flash-
oriented Coded Storage: Research Status and Future
Directions}, journal = {ACM Transactions on Storage}, year =
{2025} } @book{ostep, author = {Remzi H. Arpaci-Dusseau and
Andrea C. Arpaci-Dusseau}, title = {Operating Systems: Three
Easy Pieces}, publisher = {Arpaci-Dusseau Books}, year = {2018}
}

11:14 AM

Here are your three articles in that format:

bibtex

```
@inproceedings{weng2009hybrid,  
  author      = {Chuliang Weng and Zhigang Wang and  
Minglu Li and Xinda Lu},  
  title       = {The Hybrid Scheduling Framework  
for Virtual Machine Systems},  
  booktitle   = {Proceedings of the 2009 ACM  
SIGPLAN/SIGOPS International Conference on  
Virtual Execution Environments (VEE '09)},  
  year        = {2009}  
}
```

```
@inproceedings{zhao2017irs,  
  author      = {Yong Zhao and Kun Suo and Luwei  
Cheng and Jia Rao},  
  title       = {Scheduler Activations for  
Interference-Resilient SMP Virtual Machine  
Scheduling},  
  booktitle   = {Proceedings of the 18th ACM/IFIP/  
USENIX International Middleware Conference  
(Middleware '17)},
```

```
    year      = {2017}
}

@inproceedings{teguia2026insideout,
  author      = {Dufy Teguaia and Louis Duval and
Teo Pisenti and Kahina Lazri and Daniel Hagimont
and Thomas Pasquier and Renaud Lachaize and Alain
Tchana},
  title       = {Inside Out: A Paradigm Shift In
Live VM Introspection},
  booktitle   = {Proceedings of the 16th USENIX
Symposium on Operating Systems Design and
Implementation (OSDI '26)},
  year        = {2026},
  note        = {To appear}
}
```

Put these in your `ijcai11.bib` file. Then cite them in your text as `\cite{weng2009hybrid}`, `\cite{zhao2017irs}`, and `\cite{teguia2026insideout}`.

it is 2 articles ... the goodkit and this : Chuliang Weng, Zhigang Wang, Minglu Li, and Xinda Lu. 2009. The hybrid scheduling framework for virtual machine systems. In Proceedings of the 2009 ACM SIGPLAN/SIGOPS international conference on Virtual execution environments (VEE '09). Association for Computing Machinery, New York, NY, USA, 111–120. <https://doi.org/10.1145/1508293.1508309>

11:28 AM

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